

The Geometry of Coupled Quintessence: Parameter Sweeps, Fisher Geometry, and Packing Analysis of a Dark-Energy–Dark-Matter Interaction Model

A Consolidated Simulation and Information-Geometry Report

Computational cosmology working report

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Contents

- 1. Introduction and Motivation
- 2. The Model: Coupled Quintessence as a Dynamical System
- 3. Simulation Engine Design
- 4. Background Parameter Sweeps
- 5. Growth Sweeps: The Fifth-Force Observable
- 6. The Effective (Phantom) Layer
- 7. Fisher Geometry of the Model Manifold
- 8. Packing Geometry: Distinguishability and Optimal Covering
- Appendix A. Machine-Readable Manifold Graph
- 9. Conclusions and Caveats
- References

1. Introduction and Motivation

Recent baryon acoustic oscillation measurements from the Dark Energy Spectroscopic Instrument (DESI), combined with CMB and supernova data, hint that the dark energy equation of state may deviate from the cosmological constant value $w = -1$, with a combined preference in the Chevallier–Polarski–Linder (CPL) plane for $w_0 \approx -0.9$ and a negative time evolution $w_a < 0$ ^{[1][2]}. This report consolidates a numerical investigation of one candidate physics class—coupled quintessence, in which a scalar field ϕ rolls in an exponential potential while exchanging energy with pressureless dark matter^[3]—and reframes the parameter constraints as problems in information geometry.

The investigation proceeded in six stages, each documented below: (i) construction of a background expansion and linear-growth integration engine; (ii) parameter sweeps over the potential slope λ and coupling β ; (iii) an effective CPL layer reaching the phantom regime $w_0 < -1$ that canonical fields cannot access; (iv) the Fisher (Mahalanobis) geometry of the model manifold, including the geodesic distance from the canonical model family to the DESI-preferred region; (v) the intrinsic curvature of that manifold; and (vi) a packing-geometry analysis quantifying how many statistically distinguishable cosmologies the model family contains.

2. The Model: Coupled Quintessence as a Dynamical System

2.1 Background equations

We adopt the standard dimensionless autonomous formulation for a scalar field with exponential potential $V(\phi) = V_0 e^{-\lambda\phi/M_{\text{Pl}}}$ conformally coupled to dark matter with strength $\beta^{[4]}$. With $x \equiv \dot{\phi}/(\sqrt{6}H M_{\text{Pl}})$, $y \equiv \sqrt{V/(3H^2 M_{\text{Pl}}^2)}$, and $z \equiv \Omega_r$ the radiation fraction, evolution in e-folds $N = \ln a$ obeys

$$\begin{aligned} x' &= -3x + \sqrt{\frac{3}{2}}\lambda y^2 + \frac{3}{2}x \left(1 + x^2 - y^2 + \frac{z}{3}\right) - \sqrt{\frac{3}{2}}\beta \Omega_m, \\ y' &= -\sqrt{\frac{3}{2}}\lambda xy + \frac{3}{2}y \left(1 + x^2 - y^2 + \frac{z}{3}\right), \\ z' &= z \left(-1 + 3x^2 - 3y^2 + z\right), \end{aligned}$$

with $\Omega_m = 1 - x^2 - y^2 - z$, dark energy fraction $\Omega_\phi = x^2 + y^2$, and field equation of(1) state

$$w_\phi = \frac{x^2 - y^2}{x^2 + y^2} \geq -1,$$

where the inequality is saturated only for a vanishing kinetic term—canonical fields(2) cannot cross $w = -1$.

2.2 Linear growth with a fifth force

Sub-horizon dark matter perturbations δ_m obey

$$\delta_m'' + \left[1 + \frac{1}{2}(1 - 3w_{\text{eff}})\right] \delta_m' - \frac{3}{2}\Omega_m \frac{G_{\text{eff}}}{G} \delta_m = 0, \quad \frac{G_{\text{eff}}}{G} = 1 + 2\beta^2,$$

where the enhancement $1 + 2\beta^2$ is the scalar-mediated fifth force between dark matter(3) particles. We track $f = d \ln \delta_m / d \ln a$ and $f\sigma_8(z)$.

3. Simulation Engine Design

3.1 Failure of backward integration

An initial implementation integrated the system *backward* from present-day boundary conditions $(w_0, \Omega_{\phi,0})$. A 384-run diagnostic sweep showed that *every* such model violates

early-universe bounds catastrophically, with $\Omega_\phi(z \approx 1100) \approx 0.94$ regardless of parameters. This is the well-known attractor problem: generic late-time conditions do not lie on the early-time scaling trajectory, and thawing quintessence is unstable under time reversal. Geometrically, the viable models form an exponentially thin separatrix of the flow.

Design principle. Thawing quintessence must be integrated *forward* from early-universe initial conditions, with the present-day density targeted by a shooting method.

3.2 Forward engine and shooting

The production engine starts at $N_i = -14$ ($z \sim 10^6$) in radiation domination with correct matter–radiation balance, $x_i = 0$, $y_i = \sqrt{\Omega_{\phi,i}}$, and matter-era perturbation initial conditions. For each (λ, β) , a bisection on $\Omega_{\phi,i}$ pins $\Omega_\phi(0) = 0.685 \pm 5 \times 10^{-4}$. The required initial densities are $\Omega_{\phi,i} \sim 10^{-21}$, making the fine-tuning of thawing models explicit and quantified rather than assumed. A Λ CDM control run ($\lambda \rightarrow 0$, $\beta = 0$) calibrates all growth observables; because the sub-horizon growth equation systematically underestimates f in the radiation era, all $f\sigma_8$ results are reported as *ratios to the control*.

4. Background Parameter Sweeps

With $\Omega_{\phi,0}$ pinned, 90 models over $\lambda \in [0.1, 1.5]$ and $\beta \in [0, 0.05]$ were integrated. All satisfy the null energy condition and the early-dark-energy bound $\Omega_\phi(z \approx 1100) < 0.02$ by construction. The results show a clean degeneracy structure:

Table 1 Background sweep means over β (coupling negligible at background level)

λ	w_0	w_a	λ	w_0	w_a
0.1	-0.999	-0.002	0.9	-0.888	-0.171
0.3	-0.990	-0.022	1.1	-0.828	-0.241
0.5	-0.969	-0.058	1.3	-0.754	-0.312
0.7	-0.935	-0.109	1.5	-0.662	-0.378

The potential slope λ alone selects the position on the thawing track in the (w_0, w_a) plane; $\beta \leq 0.05$ is invisible at background level. The track passes above the DESI-preferred region, with best overlap at $\lambda \approx 0.8$ – 1.0 near the 2σ boundary (Figure 1).

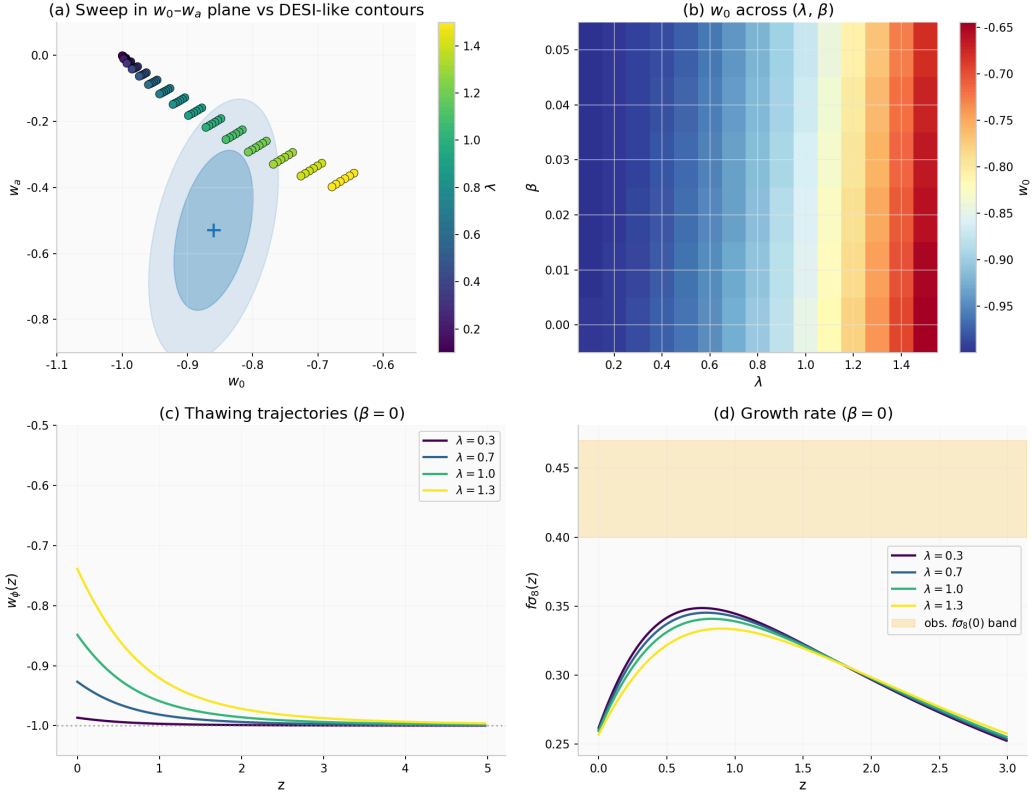


Figure 1 Background sweep: (a) thawing track vs DESI-like contours in the w_0 – w_a plane; (b) w_0 across the (λ, β) plane; (c) thawing trajectories $w_\phi(z)$; (d) growth rate $f\sigma_8(z)$ with the observational band.

5. Growth Sweeps: The Fifth-Force Observable

Extending the coupling range to $\beta \in [0, 0.2]$ (135 models) exposes where the coupling lives: the growth sector. Relative to the Λ CDM control, $f\sigma_8(z=0)$ rises by +1% at $\beta = 0.05$ and +7% at $\beta = 0.2$, scaling as $G_{\text{eff}}/G = 1 + 2\beta^2$, while varying λ changes it by only -2% . The two parameters are therefore observationally orthogonal: λ sets the expansion history, β sets structure growth—the degeneracy-breaking that joint BAO+RSD analyses exploit.

The fifth-force signature is redshift-dependent: the ratio $f\sigma_8(z)/f\sigma_8^{\Lambda\text{CDM}}(z)$ peaks near $z \approx 0.7$ – 0.9 (reaching ~ 1.4 at $\beta = 0.2$ in the sub-horizon approximation), because enhancement accumulates over the matter era while late-time dark-energy friction partially cancels it at $z = 0$ (Figure 2). This shape is a genuine discriminator against a simple σ_8 rescaling.

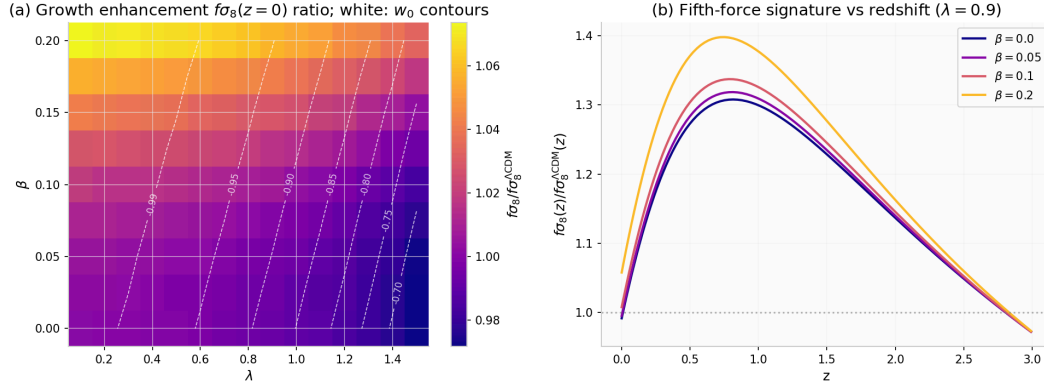


Figure 2 (a) Growth enhancement $f\sigma_8(0)/f\sigma_8^{\Lambda\text{CDM}}(0)$ across (λ, β) with w_0 contours; (b) redshift dependence of the fifth-force signature at $\lambda = 0.9$.

6. The Effective (Phantom) Layer

Because $w_\phi \geq -1$ for canonical fields (Eq. 2), the region $w_0 < -1$ was probed with an effective CPL background $w(a) = w_0 + w_a(1 - a)$, retaining the β -dependent growth equation. A sweep of 1,250 models over $w_0 \in [-1.2, -0.6]$, $w_a \in [-1.2, 0.3]$ scored against a DESI-like posterior (χ^2 Mahalanobis distance) finds:

- The posterior minimum ($w_0 \approx -0.85$, $w_a \approx -0.51$, $\chi^2 \approx 0.06$) sits at $w_0 > -1$ but with strongly negative w_a —a trajectory that *crossed* $w = -1$ from below at higher redshift (quintom-like behavior).
- The canonical thawing track only clips the upper 2σ edge; its closest point carries $\chi^2 \approx 4.8$.
- The phantom-side χ^2 landscape is flat: the data preference concerns the *crossing dynamics*, not sitting below -1 today.
- The best effective models have ΛCDM -like $f\sigma_8(0)$ but distinguishable $f\sigma_8(z)$ shapes above $z \approx 0.5$ (Figure 3).

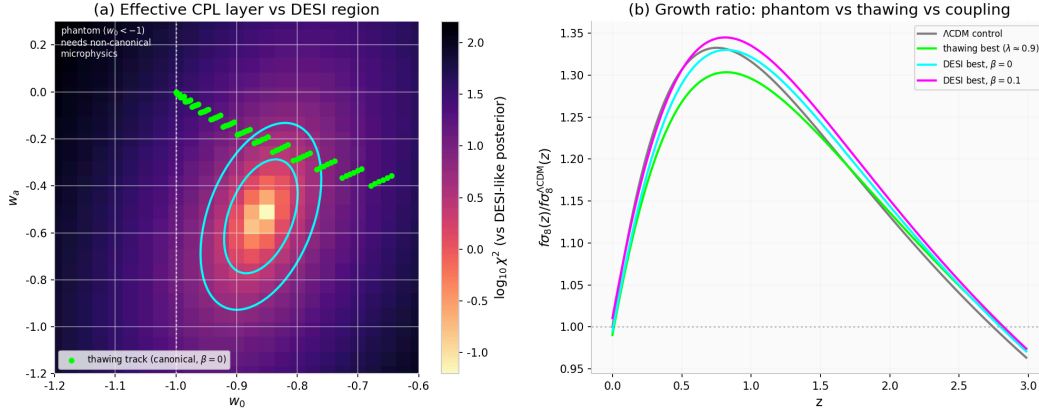


Figure 3 (a) $\log_{10} \chi^2$ of the effective CPL layer vs the DESI-like posterior, with the canonical thawing track overlaid; (b) growth ratios for the thawing best, DESI best, and DESI best with coupling.

7. Fisher Geometry of the Model Manifold

7.1 Setup

Treat the observable vector $\Theta = (w_0, w_a, f\sigma_8)$ as a point in a space endowed with the Fisher metric from the data covariance C (DESI-like w_0 – w_a covariance plus a 2% RSD growth error). The parameter manifold (λ, β) inherits the pullback metric

$$g_{ij} = (J^T C^{-1} J)_{ij}, \quad J = \frac{\partial \Theta}{\partial (\lambda, \beta)}.$$

The ellipse field of g (Figure 4a) is anisotropic (median weak/strong eigenvalue ratio $\approx (4) 0.08$) but nowhere flat once growth data is included.

7.2 The geodesic problem and its solution

Result (geodesic distance). The closest canonical thawing model to the DESI mean is $\lambda^* = 1.10$, i.e. $(w_0, w_a) = (-0.816, -0.226)$, at a Mahalanobis distance of **1.94 σ** . The geodesic direction (whitened) is $(-0.56, -0.83, 0.07)$ in $(w_0, w_a, f\sigma_8)$: the data requests primarily *steeper time evolution*, not a growth modification.

Result (parameter-space lift). Lifting the geodesic to the parameter manifold gives the step $(\Delta\lambda, \Delta\beta) = (0.33, 0.63)$ —dominated by *coupling*. Minimizing over the full (λ, β) grid yields $(1.33, 0.20)$ at **0.87 σ** , inside the 1σ ellipse.

The mechanism: matter-to-dark-energy transfer makes the dark energy density dilute more slowly than any self-conserved $w \geq -1$ fluid, so the expansion history of a coupled

canonical model *mimics phantom behavior without ghost microphysics*^[5]. At fixed λ , increasing β bends the (w_0, w_a) track toward the DESI region (Figure 4b). The geodesic is clipped at the grid boundary $\beta = 0.2$, indicating even stronger coupling would fit better—though independent CMB and local-gravity bounds restrict $\beta \lesssim 0.1$ in practice.

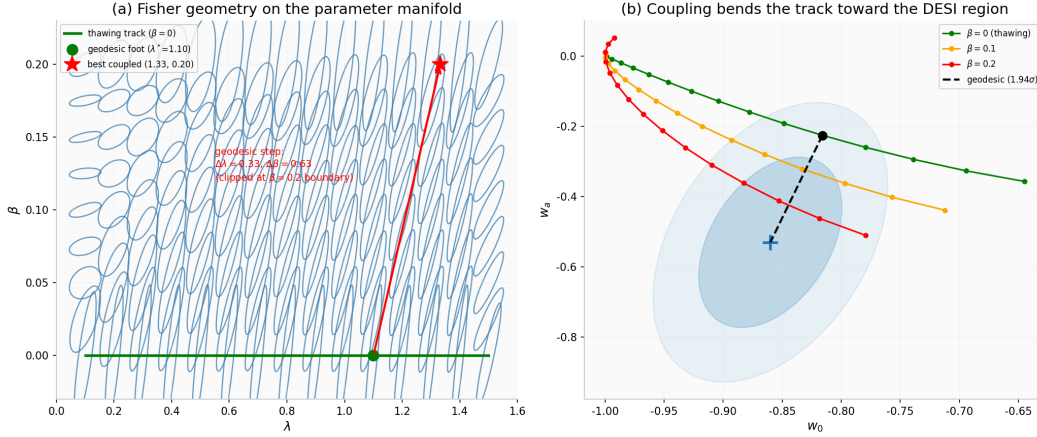


Figure 4 (a) Fisher metric ellipse field on the parameter manifold with the thawing track, geodesic foot, and best coupled model; (b) coupling bends the model track toward the DESI-preferred region.

7.3 Intrinsic curvature

Computing Christoffel symbols from second derivatives of the observable map and the Gaussian curvature $K = R_{1212}/\det g$ (Figure 5):

- **Along the geodesic path** ($\lambda \gtrsim 0.8$), $K \approx 0$. The parameterization is metrically honest where the model selection happens; linear error propagation is reliable there.
- **At $\lambda \lesssim 0.5$ the curvature estimates spike and oscillate.** There the observable map becomes nearly β -independent ($w_0 \rightarrow -1$ regardless), $\det g \rightarrow 0$, and curvature—dividing by $\det g$ —amplifies noise. This is the signature of a *caustic*: the 2D model manifold collapses toward a 1D curve at the Λ limit. Specific values in this region are unreliable, but the degeneracy itself is real.
- Mild saddle structure (K alternating sign) at intermediate λ reflects the nonlinear mapping from thawing rate to (w_0, w_a) .

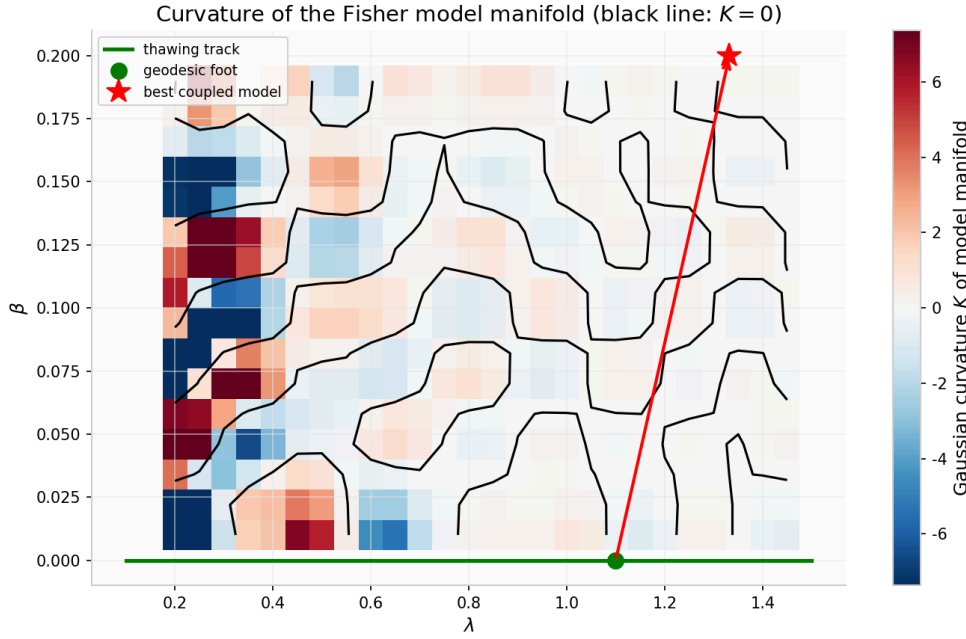


Figure 5 Gaussian curvature of the Fisher model manifold. Black contours mark $K = 0$; the geodesic path lies in a near-flat region, while the Λ limit (low λ) is a degenerate caustic.

8. Packing Geometry: Distinguishability and Optimal Covering

8.1 Model packing: four distinguishable cosmologies

Two parameter points within 1σ Mahalanobis distance are indistinguishable to the data, so counting statistically distinct cosmologies is a sphere-packing problem on the model manifold: $N = V / V_{\text{cell}}$, with volume element $\sqrt{\det g}$ and $V_{\text{cell}} = \pi$ for a 1σ disk in two dimensions. Integrating over the (λ, β) region:

Result (packing number). The Fisher volume of the model region is $V \approx 11.9 \sigma^2$, giving $N \approx 4$ **distinguishable 1σ cosmologies**. 90% of the volume lies at $\lambda \gtrsim 0.6$; the entire low- λ family compresses into $\sim 10\%$. The β direction contributes roughly uniformly (30/32/38% by band).

This is the quantitative statement of the degeneracy: theory space is large, distinguishability space tiny—and it is the same geometric event as the curvature caustic of Section 7.3, seen from the volume side.

$\pi/2 \approx 1.571$ for a square grid:

Table 2 Covering costs for full 1σ resolution of the model manifold

Grid	Covering density	Templates needed	Note
Hexagonal (optimal)	$2\pi/3\sqrt{3} \approx 1.209$	≈ 5	Kershner-optimal
Square	$\pi/2 \approx 1.571$	≈ 6	+23% vs optimal
This work (rectangular)	—	135	$\sim 25\times$ oversampled

While oversampling is harmless for cheap ODE runs, it becomes the binding constraint when each evaluation is a Boltzmann-code call. The optimal lattice is hexagonal *in the Fisher metric*—the ellipse field of [Figure 4a](#) specifies exactly how to deform the placement.

Appendix A. Machine-Readable Manifold Graph

To make the geometry accessible to downstream AI agents (and to humans who prefer graph structure over tensor calculus), the model manifold is exported as a discrete hexagonal (A_2 lattice) graph. Nodes are placed on a hexagonal lattice in (λ, β) coordinate space; edges connect node pairs whose Mahalanobis separation under the *local* Fisher metric is below 1σ — so graph adjacency literally encodes statistical distinguishability. Each node carries its full local state as a JSON payload ([Figure 6](#)): axial coordinates (q, r) , the list of topological neighbors (enabling geodesic traversal as pure graph search, with no online matrix operations), the observable vector $(w_0, w_a, f\sigma_8/f\sigma_8^{\Lambda\text{CDM}})$, the metric tensor g_{ij} and its determinant, the local Gaussian curvature K , and a singularity flag raised where $\det g$ collapses (the Λ -limit caustic of [Section 7.3](#)).

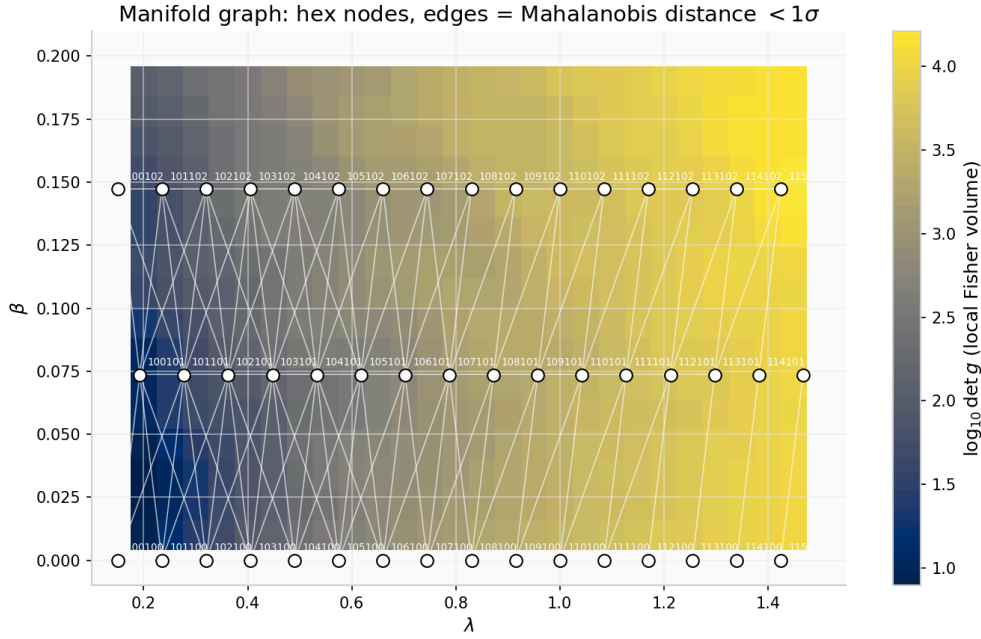


Figure 6 The 48-node hexagonal manifold graph over the Fisher-volume landscape. Edges connect nodes within 1σ Mahalanobis distance: note the dense vertical connectivity at low λ , where the metric degenerates and entire columns of parameter space become mutually indistinguishable.

The edge structure visualizes the packing result of Section 8 directly: at high λ the graph is a sparse near-chain (each node its own distinguishability cell), while at low λ columns merge into fully connected cliques — one cell swallowing many parameter points. The graph is distributed as `manifold_graph_payload.jsonl` (one JSON object per node), sized for prompt injection, multi-agent communication, or vector-store indexing. Two corrections were applied to the original encoder design: the whitening transform now uses the correct ordering $\sqrt{\Lambda} E^T$ for a symmetric metric, and the axial cell identifier is offset to avoid collisions for negative lattice coordinates.

9. Conclusions and Caveats

Conclusions. (1) Viable thawing quintessence requires forward integration with shot initial conditions; backward integration is structurally unphysical. (2) In the (λ, β) family, λ controls the expansion history and β controls growth—orthogonal observational axes. (3) The canonical thawing track misses the DESI-preferred region by 1.94σ ; the shortfall is dominantly a w_a -deficit. (4) The conformal coupling β bends the model track toward the data—a canonical route to phantom-mimicry—with a best fit at $(\lambda, \beta) \approx (1.33, 0.20)$ inside 1σ , though $\beta \gtrsim 0.1$ conflicts with independent bounds. (5) The model manifold is flat

Caveats. (i) Absolute $f\sigma_8$ values are systematically low (the Λ CDM control reads 0.26); all growth conclusions rely on ratios and shapes, and calibration requires a Boltzmann code. (ii) The DESI posterior used is a mock approximation of published contours; numbers should be refreshed with the released covariance. (iii) Coupling constraints from the CMB and local gravity were not folded into the allowed ellipse in the β direction. (iv) Curvature values near the Λ caustic are noise-amplified and indicative only.

Suggested next steps. Swap in the DESI DR2 covariance; port the background to CLASS/CAMB with a coupled-dark-energy module for calibrated $f\sigma_8$ and CMB lensing; extend the manifold with a quintom (two-field) direction and recompute the packing number.

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